

AI Workflow Automation Engine with Multi-Agent Browser Orchestration

Master Build Guide + Antigravity Prompt Pack

1. Final Project Definition

Project Title

AI Workflow Automation Engine with Multi-Agent Browser Orchestration

Project Vision

Build a scalable AI-powered workflow automation platform inspired by n8n that combines:

- AI task reasoning
- visual workflow orchestration
- browser automation
- reusable workflows
- multi-agent execution
- secure authentication
- integrations
- execution monitoring

The project must be:

- modular
 - scalable
 - maintainable
 - resume-ready
 - structured like a real SaaS platform
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2. MOST IMPORTANT DEVELOPMENT RULE

DO NOT keep adding random features.

Every feature must belong to one of these system layers:

1. Authentication Layer
2. Workflow Layer
3. Agent Layer
4. Automation Layer
5. Integration Layer
6. Execution Layer
7. Dashboard Layer
8. Storage Layer

If a feature does not fit these layers, it should NOT be added.

This keeps the project:

- clean
 - scalable
 - professional
 - maintainable
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3. Final MVP Scope

INCLUDED FEATURES

Authentication

- Login
- Signup
- JWT Authentication
- Password hashing
- Protected APIs

Workflow System

- Prompt-based workflow creation
- n8n-style workflow canvas
- Save workflow
- Reuse workflow
- Execute workflow
- Workflow history

Browser Automation

- Open websites
- Search information
- Click buttons

- Fill forms
- Extract data
- Visible/headless mode

Multi-Agent Architecture

- Planner Agent
- Browser Agent
- Execution Agent
- Verification Agent
- Summarization Agent

Integrations

- Gmail
- PDFs
- YouTube
- APIs

Dashboard

- Workflow execution history
- Logs
- Workflow analytics
- Agent execution tracking

4. EXCLUDED FEATURES (DO NOT BUILD INITIALLY)

- Mobile app
- Marketplace
- Team collaboration
- Advanced autonomous AI
- Local desktop automation
- Advanced memory/vector databases
- Complex retries/loops
- Distributed infrastructure
- Advanced scheduling engine

These can be added later.

5. Recommended Technology Stack

Layer	Technology
Frontend	React + Tailwind CSS
Workflow Canvas	React Flow
Backend	FastAPI
Browser Automation	Playwright
AI Integration	Antigravity + Gemini/OpenAI
Database	PostgreSQL
Authentication	JWT
ORM	SQLAlchemy
Realtime	WebSockets
Deployment	Docker + Render/Railway
Version Control	Git + GitHub

6. High-Level System Design

Architecture Flow

User ↓ Frontend UI ↓ Workflow API Layer ↓ AI Task Planner ↓ Workflow Orchestrator ↓ Agent Execution System ↓ Browser Automation / Integrations ↓ Execution Results ↓ Database + Logs ↓ Dashboard

7. Core System Modules

Module 1 — Authentication Layer

Responsibilities:

- User registration
- Login
- Token generation
- Session validation
- API protection

Backend Components:

- auth routes
 - JWT service
 - password hashing
 - middleware
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Module 2 — Workflow Layer

Responsibilities:

- Create workflows
- Edit workflows
- Store workflows
- Execute workflows
- Manage workflow nodes

Workflow Structure:

- Trigger Node
 - AI Node
 - Browser Node
 - Integration Node
 - Output Node
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Module 3 — Agent Layer

Responsibilities:

- Plan tasks
- Execute tasks
- Verify results
- Summarize outputs

Agents: 1. Planner Agent 2. Execution Agent 3. Browser Agent 4. Verification Agent 5. Summarization Agent

Module 4 — Browser Automation Layer

Responsibilities:

- Launch browser
- Navigate pages
- Fill forms

- Extract information
- Handle sessions

Using:

- Playwright
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Module 5 — Integration Layer

Responsibilities:

- Gmail API integration
 - PDF processing
 - YouTube summarization
 - External API communication
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Module 6 — Execution Layer

Responsibilities:

- Execute workflow steps
 - Track execution state
 - Handle workflow queue
 - Generate logs
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Module 7 — Dashboard Layer

Responsibilities:

- Workflow monitoring
 - Execution history
 - Analytics
 - Logs visualization
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Module 8 — Storage Layer

Responsibilities:

- Store workflows
- Store logs
- Store users

- Store execution history

Database:

- PostgreSQL
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8. Recommended Folder Structure

Frontend Structure

```
frontend/ | ├── src/ | ├── components/ | ├── pages/ | ├── layouts/ | ├── services/ |
          | ├── workflows/ | ├── agents/ | ├── hooks/ | ├── context/ | ├── utils/ | └── styles/ |
          | └── public/ | └── package.json
```

Backend Structure

```
backend/ | ├── app/ | ├── api/ | ├── auth/ | ├── agents/ | ├── workflows/ | ├──
browser/ | ├── integrations/ | ├── execution/ | ├── models/ | ├── database/ | ├──
services/ | ├── middleware/ | ├── utils/ | ├── requirements.txt | └── main.py
```

9. Database Design

Main Tables

users

- id
- username
- email
- hashed_password
- created_at

workflows

- id
- user_id
- workflow_name
- workflow_json
- created_at

workflow_executions

- id
- workflow_id
- execution_status
- started_at
- completed_at

logs

- id
- execution_id
- log_type
- log_message
- timestamp

integrations

- id
 - user_id
 - integration_type
 - credentials
-

10. DEVELOPMENT ROADMAP

PHASE 1 — FOUNDATION

Goal

Setup full-stack project structure.

Tasks

1. Create GitHub repository
2. Initialize React frontend
3. Initialize FastAPI backend
4. Setup PostgreSQL
5. Configure environment variables
6. Setup project architecture
7. Create base UI layout

Output

Working frontend + backend communication.

PHASE 2 — AUTHENTICATION

Goal

Build secure authentication system.

Tasks

1. Signup API
2. Login API
3. Password hashing
4. JWT token generation
5. Protected routes
6. Authentication middleware
7. Frontend auth pages

Output

Secure login system.

PHASE 3 — WORKFLOW CORE

Goal

Create workflow engine.

Tasks

1. Workflow data model
2. Create workflow APIs
3. Save workflows
4. Load workflows
5. Workflow execution APIs
6. Workflow history

Output

Core workflow system.

PHASE 4 — WORKFLOW UI

Goal

Create n8n-style workflow builder.

Tasks

1. Setup React Flow
2. Create workflow canvas
3. Create draggable nodes
4. Connect nodes
5. Save workflow JSON
6. Load workflow JSON

Output

Visual workflow editor.

PHASE 5 — AI TASK PLANNING

Goal

Convert prompts into workflow actions.

Tasks

1. Integrate Antigravity
2. Integrate Gemini/OpenAI
3. Build prompt parser
4. Create Planner Agent
5. Generate workflow steps

Output

Prompt-based workflow generation.

PHASE 6 — BROWSER AUTOMATION

Goal

Build automation engine.

Tasks

1. Setup Playwright
2. Launch browser
3. Navigate websites
4. Click elements
5. Fill forms
6. Extract webpage data
7. Support visible/headless mode

Output

Working browser automation.

PHASE 7 — MULTI-AGENT SYSTEM

Goal

Implement multi-agent orchestration.

Tasks

1. Planner Agent
2. Browser Agent
3. Execution Agent
4. Verification Agent
5. Summarization Agent
6. Agent communication flow

Output

Multi-agent execution pipeline.

PHASE 8 — INTEGRATIONS

Goal

Add external integrations.

Tasks

1. Gmail integration
2. PDF parser

3. YouTube summarizer
4. API connectors

Output

Integrated automation ecosystem.

PHASE 9 — DASHBOARD & LOGGING

Goal

Create monitoring system.

Tasks

1. Execution history
2. Workflow logs
3. Analytics basics
4. Agent activity logs

Output

Professional dashboard.

PHASE 10 — DEPLOYMENT

Goal

Deploy project professionally.

Tasks

1. Docker setup
2. Environment configs
3. Backend deployment
4. Frontend deployment
5. SSL/domain setup
6. Production testing

Output

Production-ready project.

11. ANTIGRAVITY MASTER PROMPT

Copy and use this inside Antigravity.

Project Name: AI Workflow Automation Engine with Multi-Agent Browser Orchestration

Objective: Build a scalable AI-powered workflow automation platform inspired by n8n that combines:

- workflow orchestration
- browser automation
- AI task planning
- multi-agent execution
- reusable workflows
- integrations
- secure authentication

Requirements:

1. Use React + Tailwind CSS for frontend.
2. Use FastAPI for backend.
3. Use PostgreSQL database.
4. Use React Flow for workflow canvas.
5. Use Playwright for browser automation.
6. Implement JWT authentication.
7. Create modular scalable architecture.
8. Build multi-agent system with:
 9. Planner Agent
 10. Execution Agent
 11. Browser Agent
 12. Verification Agent
 13. Summarization Agent
14. Support:
 15. prompt-based workflows
 16. visual workflow editing
 17. workflow execution
 18. workflow saving/reusing
19. Add integrations:
 20. Gmail

21. PDFs
22. YouTube
23. APIs
24. Implement workflow logs and execution tracking.
25. Use clean production-style folder structure.
26. Prioritize MVP-first development.
27. Avoid unnecessary features initially.
28. Follow scalable SaaS architecture principles.

Generate:

- architecture
 - APIs
 - database schema
 - development roadmap
 - backend services
 - frontend components
 - workflow system
 - execution engine
 - security implementation
 - deployment strategy
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12. HOW TO WORK WITH ANTIGRAVITY

IMPORTANT RULE

DO NOT ask Antigravity: “Build entire project.”

Instead:

- build module-by-module
 - verify each phase
 - test before next phase
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Correct Development Pattern

Step 1

Ask: “Generate backend folder structure and FastAPI initialization.”

Step 2

Ask: "Generate PostgreSQL database configuration."

Step 3

Ask: "Generate JWT authentication system."

Step 4

Ask: "Generate workflow models and APIs."

Build sequentially.

13. PROJECT SUCCESS STRATEGY

Your project will become strong NOT because of many features.

It becomes strong because:

- architecture is clean
- modules are separated
- workflows actually work
- browser automation is reliable
- UI is professional
- authentication is secure
- codebase is maintainable

Focus on:

- structure
- execution
- usability
- stability

NOT random feature addition.

14. FINAL MVP CHECKLIST

BEFORE ADDING NEW FEATURES

Always ask:

1. Does this fit the architecture?
2. Is this necessary for MVP?
3. Will this increase complexity too much?
4. Is the current system stable?
5. Is the feature reusable?

If answer is NO → do not add.

15. FINAL PROJECT GOAL

Build a professional AI automation platform that demonstrates:

- system design
- workflow orchestration
- AI integration
- browser automation
- backend engineering
- frontend architecture
- scalable software design

This should feel like:

- startup software
- enterprise SaaS
- modern AI automation platform

NOT a normal student CRUD project.